

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 5

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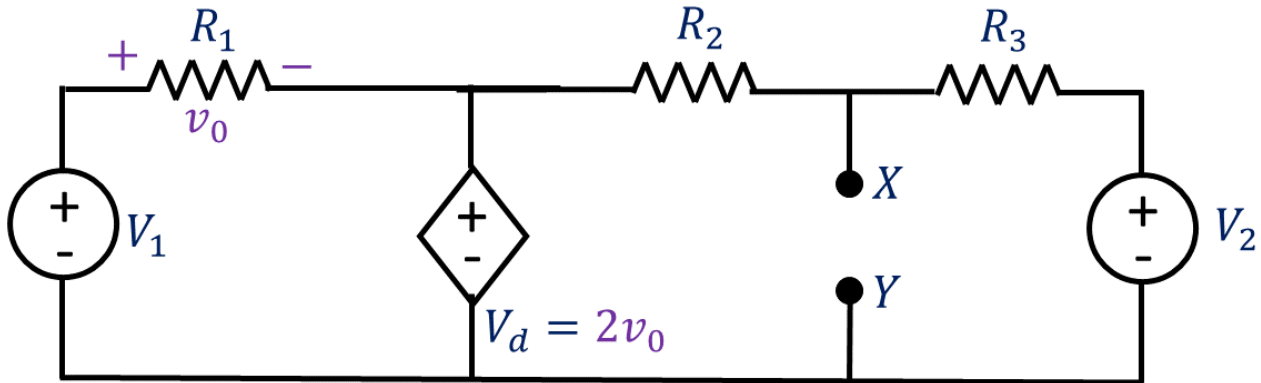


Figure 1 The network shown in the above figure has one dependent voltage source $V_d = 2 \times v_0$, two independent voltage sources $V_1 = 18V$ and $V_2 = 15V$ along with three resistors $R_1 = 1\Omega$, $R_2 = 5\Omega$ and $R_3 = 4\Omega$. A load resistance can be connected across the two accessible terminals X and Y .

1. Consider the network shown in Figure 1. Estimate the numerical value of the optimal resistive load R_L^* and the maximum power P_{max} that will be drawn by R_L^* from this network.
2. We need to design a code converter from the Excess-3 code to the 2-out-of-5 code shown in the figure below. Find minimal SOP expressions and minimal POS expressions for each output.
3. We need to design a code converter from the BCD code to the Ringtail code shown in the figure below. Find minimal SOP expressions and minimal POS expressions for each output.

Decimal	2-out-of-5				
0	1	1	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	0
7	1	0	0	0	1
8	1	0	0	1	0
9	1	0	1	0	0

Decimal	Ringtail				
0	0	0	0	0	0
1	0	0	0	0	1
2	0	0	0	1	1
3	0	0	1	1	1
4	0	1	1	1	1
5	1	1	1	1	1
6	1	1	1	1	0
7	1	1	1	0	0
8	1	1	0	0	0
9	1	0	0	0	0